

Computational evidence toward Artin holomorphy for the minimal even icosahedral representation

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22 September 2026

Abstract

Let ρ be the even two-dimensional Artin representation of conductor 1951 attached to the Doud–Moore totally real A_5 quintic of discriminant 1951^4 . If Artin’s holomorphy conjecture holds, ρ corresponds to a Maass cusp form of level 1951, nebentypus of order 5, and Laplace eigenvalue $1/4$. We determine the $\mathrm{SL}_2(\mathbb{F}_5)$ -lift of ρ from class-field data of the sextic resolvent, separate the two classes of 5-cycles by Dokchitser resolvents and local inertia, and compute the algebraic root number

$$W(\rho) = -0.450140208385 - 0.892957889710i.$$

The resulting Euler product matches a numerical Hejhal stream at every prime the stream resolves, satisfies a certified functional-equation residual $|R(y)| \leq 3.3 \cdot 10^{-133}$ at 26 points, and satisfies 37 certified twisted residuals of size at most $9.4 \cdot 10^{-52}$. Assuming entirety, Turing’s method at a common cutoff $T_0 = 146$ gives exactly 277 zeros with $0 < \Im s \leq 146$ and exactly 276 zeros with $0 < -\Im s \leq 146$, all witnessed by certified sign changes on the critical line. None of this proves Artin’s conjecture for ρ .

MSC 2020. 11F80, 11F72, 11R32, 11Y35.

Keywords. Artin holomorphy, even icosahedral representations, Maass forms, Hejhal’s algorithm, Turing’s method.

1 What the conjecture is, and what this note is not

Artin’s holomorphy conjecture asserts that if $\rho: \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \mathrm{GL}_n(\mathbb{C})$ is continuous, irreducible and nontrivial, then $L(s, \rho)$ is entire. Automorphy of ρ implies holomorphy, by Godement–Jacquet; the converse is false in general. For irreducible two-dimensional representations over $\mathbb{Q}$, Booker showed that the existence of only finitely many poles already forces automorphy [2]. Solvable image is known (Langlands–Tunnell [8]). Odd icosahedral image is modular over $\mathbb{Q}$ as a weight-one holomorphic form, by Serre’s conjecture (Khare–Wintenberger [7]) together with the weight-one lifting theorems [1]. The remaining two-dimensional line over $\mathbb{Q}$ is even icosahedral image: a representation ρ with projective image A_5 and $\det \rho(c) = +1$ should be a Maass cusp form of eigenvalue $1/4$.

Doud–Moore constructed six such representations of prime conductor $p \equiv 1 \pmod{5}$, and proved by targeted Hunter search that 1951 is the smallest prime conductor of an even icosahedral representation [6]. Booker–Lee–Strömbergsson, assuming Artin holomorphy, classified even two-dimensional Artin representations of conductor at most 2862 and confirmed that this same representation is the first even icosahedral example in that range [4]. That is the object of this note:

$$f_5 = x^5 - x^4 - 780x^3 - 1795x^2 + 3106x + 344, \quad \mathrm{disc} K = 1951^4, \quad \mathrm{Gal}(f_5) \cong A_5.$$

The associated ρ is even, of conductor 1951, with nebentypus χ the even Dirichlet character modulo 1951 of order 5 (Conrey index 780).

Remark 1. A finite computation cannot prove that $L(s, \rho)$ is entire, nor that $a_p(\varphi) = \text{tr } \rho(\text{Frob}_p)$ at every prime for a Maass form φ . The results below are certificates about explicit fields, explicit Euler factors, explicit residuals, and a *conditional* zero count. They are not a proof of Artin holomorphy for ρ .

2 The lift to $\text{SL}_2(\mathbb{F}_5)$

Write χ for the order-5 character above and set $\rho_0 = \rho \otimes \chi^{-3}$. Then $\det \rho_0 = 1$, so ρ_0 takes values in $\text{SL}_2(\mathbb{F}_5)$. At unramified p ,

$$a_p = \chi(p)^3 t_p, \quad t_p = \text{tr } \rho_0(\text{Frob}_p) \in \{0, \pm 1, \pm 2, \pm \varphi, \pm(\varphi - 1)\},$$

where $\varphi = (1 + \sqrt{5})/2$. The projective A_5 -class of Frob_p fixes $|t_p|$ up to the $5A/5B$ distinction. The sign of t_p lives in the kernel field N_0 of ρ_0 , a quadratic extension of the A_5 -closure, unramified at every finite prime and ramified at every real place (because $\rho_0(c) = -I$).

Let K_6 be the totally real sextic resolvent

$$y^6 - 75y^4 - 70y^3 + 826y^2 - 210y - 625,$$

of discriminant 1951^4 and Galois group $\text{PSL}_2(\mathbb{F}_5)$. PARI's `bnfcertify` gives $\text{Cl}(K_6) = C_2$ and $\text{Cl}^+(K_6) = C_4 \times C_2 \times C_2$. Exactly one index-4 quotient of the narrow class group is cyclic and ramified at all six real places. Its class field $K_{24} = N_0^H$, cut by a unipotent $C_5 \leq \text{SL}_2(\mathbb{F}_5)$, is totally complex of discriminant 1951^{16} . Factorisation of p in K_{24} is the order of $\rho_0(\text{Frob}_p)$:

quintic type	K_{24} type	order in $\text{SL}_2(\mathbb{F}_5)$	t_p
1^5	$1^{24}/2^{12}$	$1/2$	$+2/-2$
$2+2+1$	4^6	4	0
$3+1+1$	$3^8/6^4$	$3/6$	$-1/+1$
5	$1^4 5^4/2^2 10^2$	$5/10$	$\S 3$

Among the 2261 primes below 20 000, the four rows contain 27, 575, 769 and 890 primes. Chebotarev predicts about 38, 565, 754 and 904.

3 The classes $5A$ and $5B$

All 5-cycles are conjugate in S_5 . In A_5 they split into two classes. Dokchitser resolvents $\Gamma_A, \Gamma_B \in \mathbb{Z}[X]$ separate them: at each of the 890 such primes below 20 000, three choices of auxiliary polynomial never disagree once at least one resolvent is conclusive modulo p .

Which class has $|t| = \varphi - 1$ is local at 1951. The roots of f_5 lie in $\mathbb{Q}_{1951}(\pi)$ with $\pi^5 = -f_5(1561)$. An inertia element g restricting to the Teichmüller twist $\pi \mapsto \omega(3)^{390}\pi$ has $\chi(g) = \chi(3) = \zeta_5^2$. Hence $\rho(g)$ has eigenvalues 1 and ζ_5^2 , so $\rho_0(g)$ has eigenvalues ζ_5 and ζ_5^4 and trace $\varphi - 1$. The same element lies in class A of the resolvents. Thus:

class	order 5	order 10
A	$t = \varphi - 1$	$t = 1 - \varphi$
B	$t = -\varphi$	$t = +\varphi$

The Galois conjugate $\rho_0^\sigma \otimes \chi^3$, $\sigma(\sqrt{5}) = -\sqrt{5}$, has no inertia-fixed line at 1951 and conductor 1951^2 .

Every prime of K_{24} above 1951 has residue degree 1, so the ramified Euler factor is $a_{1951} = \chi(877)^{-3} = \zeta_5^2$. The Artin root number is

$$W(\rho) = -\frac{\tau(\chi) \overline{a_{1951}}}{\sqrt{1951}} = -0.450140208385 - 0.892957889710i. \quad (1)$$

The Fricke phase of the putative Maass form is the algebraic constant $\lambda_N = -W(\rho)$. This value is frozen throughout. Least-squares fits of a numerical phase (“ λ_{opt} ”) are discarded.

4 Comparison with a numerical Maass stream

A float64 Hejhal expansion at level 1951, odd sector, truncation $M = 8000$, was compared with the Galois Euler factors at every prime $p < 5639$. Sign and magnitude agree at all 738 such primes, including the nine completely split primes (Frob = +1) and $p = 5279$ (Frob = −1, $a_p = -2\chi(p)^3$). Agreement is $8 \cdot 10^{-6}$ for $p < 1000$ and $2.6 \cdot 10^{-5}$ for $p < 2000$. Beyond that the numerical coefficients degrade; the comparison stops at $p = 5639$, where the Hejhal error reaches 0.4.

This is a numerical statement. It becomes a theorem for those primes only when the Hejhal coefficients carry certified enclosures. A double-double comparison against the $M = 12\,000$ stream remains to be run on the machine that holds those coefficient files.

5 Functional-equation residuals

Brauer meromorphy and the Artin functional equation give

$$R(y) := A(y^{-1}) - W(\rho) y \overline{A(y)} = \sum_{\text{poles } \varrho} \text{Res}_{s=\varrho}(\Lambda(s) y^s),$$

where A is the incomplete- K_0 theta series of $L(s, \rho)$. Thus $R \equiv 0$ if and only if $L(s, \rho)$ is entire.

Arb ball arithmetic at 1600 bits, with exact algebraic coefficients through $n \leq 8981$ and a tail bound $|a_n| \leq d(n)$, yields

$$|R(y)| \leq 3.3 \cdot 10^{-133} \quad \text{at } y = 0.25, 0.28, \dots, 1.00$$

(26 points), and the same order at the reciprocal points. A simple pole on the critical line at height 10 would need residue $\leq 5 \cdot 10^{-128}$; at height 180, residue $\leq 10^{-12}$. There is no lower bound on a residue, and several poles could cancel on a finite grid. The residual is therefore evidence, not a proof of entirety.

A float64 survey of all 661 primitive twists of conductor $q \leq 60$ ($q \not\equiv 2 \pmod{4}$) produces a best-fit constant of modulus 1 to 10^{-15} , matching the local root-number formula

$$W_\psi = W(\rho) \psi(-1) \chi(q) \psi(1951) \tau(\psi)^2 / q$$

to $1.7 \cdot 10^{-14}$, with largest relative residual $4.5 \cdot 10^{-14}$. All 37 primitive twists of conductor $q \leq 13$ have certified residual at most $9.4 \cdot 10^{-52}$ (Arb, 50 digits). Weil’s converse theorem would require entirety of infinitely many twists, with growth. Finite twists are necessary conditions, not a converse.

Booker’s finite-height argument [2] cannot reach this ρ : every entire L -function that contains the block of ρ has conductor at least 1951^8 .

6 A test-function certificate, and its limit

Feeding the exact Artin coefficients ($n \leq 12\,000$) and the algebraic phase $\lambda_N = -W(\rho)$ into a Booker–Strömbergsson–Venkatesh style jump computation on the Ford sides of $\Gamma_0(1951)$ [5] gives

$$Q_0 \leq 3.16 \cdot 10^{-22}, \quad Q_1 \leq 7.39 \cdot 10^{-22},$$

dominated by a priori evaluation-error bars of size about 10^{-13} , not by a residual automorphy defect. An independent assembler returns $Q_0 \leq 4.43 \cdot 10^{-21}$ and $Q_1 \leq 1.25 \cdot 10^{-20}$. The formal output of the existing Sobolev recipe at $\delta = 0.75$ is $|\lambda - 1/4| \leq 6.1 \cdot 10^{-10}$.

Two caveats are essential. First, the constants in that recipe were derived for $\mathrm{PSL}_2(\mathbb{Z})$ and have not been re-proved for $\Gamma_0(N)$ with nebentypus; the number $6.1 \cdot 10^{-10}$ is a figure of merit, not an enclosure. Second, the test function is the Artin series itself. Small jumps mean that this Euler product is almost automorphic on the Ford sides. They do not construct an independent eigenfunction and then identify it with ρ .

7 Quasi-modes versus isolation

If u is simultaneously an ε_0 -eigenfunction of Δ and an ε_p -eigenfunction of T_p for $p \in S$, then some joint Hecke–Maass newform φ satisfies

$$|\lambda(\varphi) - \tfrac{1}{4}| \leq \sqrt{2}\varepsilon_0, \quad |\lambda_\varphi(p) - a_p| \leq \sqrt{2|S|}\varepsilon_p \quad (p \in S).$$

Spectral isolation is not used for this finite- S statement. Isolation is still required for uniqueness near $\lambda = 1/4$ and for identification at every prime.

A Monte Carlo sampler of $\|(T_p - a_p)F\|/\|F\|$ on 800 hyperbolic points of the Ford domain returns $3.5\text{--}4.2 \cdot 10^{-15}$ for the Artin series at $p = 2, 3, 5, 7, 11, 13$, and $O(1)$ for the swapped $5A/5B$ control (conductor 1951^2). That is a float64 floor on a thin sample, not an L^2 bound. Projected figures such as $\varepsilon_p \sim 10^{-11}$ are withdrawn until a two-dimensional certified integral exists.

8 Conditional zeros to height 146

Assume $L(s, \rho)$ is entire, so that the functional equation may be used on the negative half-line of the incomplete- K_0 integral. Let

$$Z(t) = W(\rho)^{-1/2} \Lambda(\tfrac{1}{2} + it).$$

A trapezoid rule of step $h = 1/60$ in a strip of half-width 1.3, with Trefethen–Weideman error bounds [10] and 1484-bit Arb nodes, produces certified signs of Z at all 15 001 points of a 0.02 -grid on $[-150, 150]$. The argument of Λ along $[1/2, 3]$ is controlled, and $|L(\sigma) - 1| \leq \zeta(3)^2 - 1 < 1$ for $\sigma \geq 3$, so there are no real zeros and

$$S(0) = \frac{\arg W(\rho)}{2\pi} = -0.3243129998.$$

Palojärvi–Zhao Theorem 2.1 [9], applied at a common cutoff $T_0 = 146$ with window length 4, then excludes hidden zeros below that cutoff on both sides. Combined with the certified sign changes one obtains:

Proposition 2 (conditional). *Assume $L(s, \rho)$ is entire. Then $L(s, \rho)$ has exactly 277 zeros with $0 < \Im s \leq 146$ and exactly 276 zeros with $0 < -\Im s \leq 146$. Each is witnessed by a certified sign change of Z on an interval of length 0.02. There is one further certified sign change in $[-146.58, -146.56]$, outside the symmetric window.*

The two sides are not complex conjugates of a real L -function: χ is complex, so the counts need not agree. An earlier run that kept $T_0 = 147$ on the negative side counted the extra zero at -146.57 and cleared Turing by only 0.087 . The symmetric cutoff is the one we adopt.

“All simple” here means one sign change per cell and no missing zeros in the window. It is not a multiplicity theorem.

9 What remains

1. A double-double comparison of the Galois table with the $M = 12\,000$ Hejhal stream, on the machine that holds those coefficients.
2. A certified L^2 bound for $(T_p - a_p)F$ on $\mathfrak{F}^+ \cup W_{1951}\mathfrak{F}^+$, beginning at $p = 2$.
3. A proof of the Sobolev constants in the jump-to-eigenvalue recipe for $\Gamma_0(N)$ with nebentypus, or a replacement of that recipe.
4. Spectral isolation near $\lambda = 1/4$ in the odd χ -space, if uniqueness is desired.

None of these, and no longer list of residuals, proves Artin’s conjecture for ρ . That statement is either “ $L(s, \rho)$ is entire”, or “ $a_p(\varphi) = \text{tr } \rho(\text{Frob}_p)$ at every prime for a Maass form φ of eigenvalue $1/4$ ”. No finite computation gives either.

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